

2. Review & Analyze Requirements

2.1 Introduction

Requirement Analysis is one of the most important phases in the Software Testing Life Cycle (STLC). Before designing test scenarios, test cases, or automation scripts, it is essential to thoroughly understand the business requirements, application functionality, user expectations, and system behavior.

The purpose of requirement analysis is to identify what needs to be tested, understand the application's business flow, determine the testing scope, identify risks, and ensure that all requirements can be validated through testing activities.

For this project, the Parabank Banking Application was analyzed to identify key business functionalities, user workflows, functional requirements, non-functional requirements, dependencies, constraints, and testable features. The analysis served as the foundation for creating test scenarios, test cases, automation scripts, and CI/CD integration.

2.2 Application Overview

Parabank is an online banking web application that allows customers to perform various banking activities through a web interface.

The application provides banking services such as:

- User Registration
- User Authentication (Login/Logout)
- Account Management
- Account Overview
- Open New Account
- Fund Transfer
- Bill Payment
- Transaction Search
- Loan Request
- Contact Information Update

The application simulates real-world banking operations and is suitable for automation testing and CI/CD implementation.

2.3 Requirement Gathering Sources

The application requirements were collected and analyzed from the following sources:

1. Application Functionalities

The functionality available within the Parabank application was reviewed to understand business operations and user workflows.

2. User Interface Analysis

Application pages, forms, buttons, links, messages, and workflows were analyzed to identify testable elements.

3. Business Flow Review

Banking operations performed by end users were studied to understand how different modules interact with each other.

4. Application Navigation

Page navigation and user journeys were analyzed to determine dependencies between different modules.

2.4 Functional Requirement Analysis

The following functional requirements were identified during analysis.

Module 1: User Registration

Requirement Description

The application should allow new customers to register successfully by providing valid personal and account information.

Functional Requirements

- User should be able to access registration page.
- User should be able to enter customer details.
- User should be able to create username and password.
- System should validate mandatory fields.
- System should prevent duplicate usernames.
- Successful registration message should be displayed.

Testability Status

Testable

Module 2: User Login

Requirement Description

Registered users should be able to login using valid credentials.

Functional Requirements

- Login page should be accessible.
- User should enter valid username.
- User should enter valid password.
- System should authenticate user.
- User should be redirected to account dashboard.
- Error message should be displayed for invalid credentials.

Testability Status

Testable

Module 3: Account Overview

Requirement Description

Users should be able to view account information after successful login.

Functional Requirements

- Account list should be displayed.
- Account numbers should be visible.
- Account balances should be visible.
- Account summary should be accurate.
- Account details page should be accessible.

Testability Status

Testable

Module 4: Open New Account

Requirement Description

Users should be able to create new bank accounts.

Functional Requirements

- Open Account page should be accessible.
- Savings account creation should be supported.
- Checking account creation should be supported.
- New account should appear in account overview.
- Confirmation message should be displayed.

Testability Status

Testable

Module 5: Transfer Funds

Requirement Description

Users should be able to transfer money between accounts.

Functional Requirements

- Source account selection.
- Destination account selection.
- Transfer amount entry.
- Balance validation.
- Transfer confirmation.
- Transaction recording.

Testability Status

Testable

Module 6: Bill Payment

Requirement Description

Users should be able to pay bills using account balance.

Functional Requirements

- Enter payee information.
- Enter account information.
- Enter payment amount.

- Submit payment request.
- Generate confirmation message.
- Update transaction history.

Testability Status

Testable

Module 7: Find Transactions

Requirement Description

Users should be able to search transactions using different search criteria.

Functional Requirements

- Search by transaction ID.
- Search by transaction date.
- Search by date range.
- Search by amount.
- Display accurate results.

Testability Status

Testable

Module 8: Request Loan

Requirement Description

Users should be able to request loans based on account eligibility.

Functional Requirements

- Enter loan amount.
- Enter down payment.
- Submit loan request.
- Loan approval process.
- Loan rejection process.
- Confirmation message display.

Testability Status

Testable

Module 9: Update Contact Information

Requirement Description

Users should be able to update personal information.

Functional Requirements

- Update address.
- Update phone number.
- Update ZIP code.
- Save updated information.
- Persist updated information after login.

Testability Status

Testable

2.5 Non-Functional Requirement Analysis

In addition to business functionality, the following non-functional requirements were considered.

Performance

- Application pages should load within acceptable response time.
- Transactions should be processed efficiently.

Usability

- User interface should be easy to navigate.
- Error messages should be clear and meaningful.

Reliability

- Application should consistently perform expected operations.

Security

- User credentials should remain protected.
- Unauthorized access should be prevented.

Availability

- Application should remain accessible during testing activities.
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2.6 Requirement Traceability Considerations

Each identified requirement should be traceable throughout the testing process.

The following artifacts should be mapped:

Requirement → Test Scenario → Test Case → Automation Script → Test Execution Result → Defect

This traceability ensures complete requirement coverage and helps identify any missing test coverage.

2.7 Risk Analysis

The following risks were identified during requirement analysis.

Risk ID	Risk Description	Impact
R001	Application downtime during execution	High
R002	Dynamic element identification issues	Medium
R003	Test data inconsistency	Medium
R004	Changes in application UI	High
R005	Browser compatibility issues	Medium
R006	Network latency affecting execution	Medium

2.8 Assumptions

The following assumptions were considered:

- Application is available throughout testing.
- Test users are available.
- Browser and driver versions are compatible.
- Test environment remains stable.
- Application functionality remains unchanged during execution.

2.9 Requirement Analysis Outcome

After analyzing the Parabank application requirements, the following outcomes were achieved:

- Business workflows identified.
- Functional requirements documented.
- Non-functional requirements identified.
- Testable modules finalized.
- Risks identified.
- Automation scope defined.
- Test scenario preparation initiated.

2.10 Conclusion

Requirement analysis provided a comprehensive understanding of the Parabank Banking Application and established a strong foundation for subsequent testing activities. The identified requirements, risks, assumptions, and business workflows were used to design test scenarios, test cases, automation scripts, and CI/CD processes. Proper requirement analysis ensured complete requirement coverage and improved the effectiveness of the overall testing strategy.